

INTERDISCIPLINARY RESEARCH ARTICLE

Hyper-Ribbon Manifolds of Prediction Error

Universal Low-Dimensional Structure in Parameterized Models, with Ecological Fallacy Detection

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ABSTRACT

Parameterized models in science and engineering are typically evaluated by their predictive accuracy on test benchmarks. A less explored question is the *geometric structure* of the errors themselves. Here we report a systematic study of 559 independently parameterized models for predicting elastic constants of 15 metallic elements. We find that prediction errors universally occupy low-dimensional **hyper-ribbon manifolds** with effective dimensionality 1.05–1.86 in a 3-dimensional observable space — the signature of sloppy model universality. This compression persists across four decades of model development and extends to foundation machine-learning potentials (MACE-MP-0, CHGNet, Orb-v3), indicating the phenomenon is paradigm-independent.

A striking crystal-structure dichotomy governs error correlations: all body-centered cubic (BCC) metals show strong reference–prediction agreement ($r > 0.70$), while face-centered cubic (FCC) metals show near-zero correlation ($r < 0.40$). Random-effects meta-analysis across 15 elements ($N = 1,677$) reveals extreme heterogeneity ($I^2 = 98.6\%$), demonstrating that naive cross-element aggregation commits **ecological fallacy**. A direct test of the *d*-band chemistry hypothesis finds it refuted on the full sample; the dominant confounder is fitting-sample size rather than electronic structure. All methods are implemented in the open-source `atlas-distill` engine.

Keywords: sloppy models, information geometry, meta-analysis, Simpson's paradox, interatomic potentials, machine learning, manifold learning, ecological fallacy

1. Introduction

Every parameterized model in science produces prediction errors. Whether the model is a neural network for image classification, a differential equation for fluid dynamics, or an analytical potential for atomistic simulation, the errors are rarely random noise — they carry geometric structure that encodes information about the model, the data, and the underlying physics. Understanding this structure is essential for model improvement, uncertainty quantification, and the design of ensemble methods.

In this paper we study the geometry of prediction errors in a domain that sits at the intersection of physics, statistics, and machine learning: **interatomic potentials**. These are mathematical models that describe the energy of an atomic configuration as a function of nuclear positions. They are used to simulate materials at the atomic scale — predicting crystal structures, elastic properties, defect energies, and phase transitions — and serve as the workhorse of computational materials science. Thousands of such potentials have been developed over forty years, each with different functional forms, fitting procedures, and target properties.

This wealth of independently parameterized models provides an ideal natural experiment for studying error geometry.

We ask three questions that are general enough to interest any quantitative scientist:

Q1 (Geometry). Do prediction errors distribute isotropically in the space of target observables, or do they concentrate onto low-dimensional manifolds? The latter would imply that the model failures are *structured* — certain directions in observable space are well-captured while others are systematically missed.

Q2 (Heterogeneity). When multiple models are evaluated across multiple systems, can their performance be meaningfully aggregated? Or does the aggregation itself produce misleading conclusions through ecological fallacy — the statistical artifact where group-level correlations diverge from individual-level ones?

Q3 (Universality). Does the error geometry depend on the model architecture (e.g., classical analytical functions versus neural networks), or is it a property of the observable space itself?

These questions are motivated by three well-established intellectual traditions. First, **sloppy model theory**^[1]^[2] from information geometry establishes that multiparameter models generically have parameter sensitivities spanning many orders of magnitude. Only a few "stiff" parameter combinations control predictions, while many "sloppy" directions can vary freely without affecting observable quantities. The resulting *model manifold* — the set of all possible predictions — forms a hyper-ribbon in data space^[3].

Second, **Simpson's paradox**^{[4][5]} and the ecological fallacy^[6] warn that correlations computed across heterogeneous groups can invert when confounders are controlled. In materials benchmarking, element identity is a confounder: crystal structure, bonding character, and atomic size simultaneously influence both the target properties and the model's predictive accuracy. Aggregating across elements without stratification risks precisely this fallacy.

Third, **meta-analysis**^{[7][8]} provides rigorous statistical frameworks for combining effect sizes across heterogeneous studies. Random-effects models account for between-group variance, and heterogeneity statistics (I^2 , τ^2) quantify whether aggregation is appropriate at all.

Our contributions are: (i) the first demonstration that elastic constant prediction errors for 559 real potentials universally occupy hyper-ribbon manifolds with effective dimensionality 1.05–1.86; (ii) discovery of a crystal-structure-dependent dichotomy in error correlations; (iii) the first application of random-effects meta-analysis and ecological fallacy detection to multi-model benchmarking; (iv) proof that hyper-ribbon structure extends to foundation machine-learning potentials, establishing paradigm independence. All analyses are implemented in `atlas-distill`, an open-source Rust engine.

2. Theoretical Framework

2.1 Sloppy models and hyper-ribbon geometry

For a model with N parameters $\boldsymbol{\theta}$ fit to M observations, the Fisher Information Matrix $\mathbf{H}_{ij} = \langle \partial_i \partial_j \ln L \rangle$ characterizes the local curvature of the likelihood landscape. Its eigenvalue decomposition $\mathbf{H}\mathbf{v}_k = \lambda_k \mathbf{v}_k$ reveals the geometry of the model manifold.

The **participation ratio** measures the effective dimensionality of the error distribution:

$$\text{PR} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} \quad (1)$$

For isotropic data in d dimensions, $\text{PR} = d$; for data confined to a k -dimensional subspace, $\text{PR} = k$. Sloppy models exhibit approximately linear log-spacing of eigenvalues $\ln \lambda_k \approx a + b \cdot k$, the signature of Vandermonde matrix ensembles^[9]. We classify hyper-ribbon structure by three criteria: (1) monotonic eigenvalue decay (Mann-Kendall $\tau < -0.8$), (2) high log-linearity ($R^2 > 0.8$), and (3) fractional dimensionality ($\text{PR}/d < 0.9$).

2.2 Ecological fallacy in multi-system benchmarking

Given grouped data $\{(x_i, y_i, g_i)\}$ where g_i labels the group (here: chemical element), define:

- *Pooled correlation* r_{pool} across all data points
- *Within-group correlation* r_g for each group g
- *Mean within-group correlation* $\bar{r} = \sum_g n_g r_g / \sum_g n_g$

Following Kievit et al.^[10], we detect ecological fallacy when $|r_{\text{pool}} - \bar{r}| > 0.3$ and the between-group variance in both variables is substantial. Because elastic constants have inherent physical constraints that create non-zero baseline correlations, permutation testing against the empirical null distribution is required rather than testing against $\rho = 0$.

2.3 Random-effects meta-analysis

For K groups with correlations r_k and sample sizes n_k , the Fisher z-transformation is $z_k = \text{arctanh}(r_k)$ with $\text{SE}(z_k) = 1/\sqrt{n_k - 3}$. Fixed-effects weights by inverse variance; random-effects adds the

DerSimonian-Laird between-study variance^[11]:

$$w_k^* = \frac{1}{\text{SE}(z_k)^2 + \tau^2}, \quad \tau^2 = \max\left(0, \frac{Q - (K - 1)}{\sum w_k - \sum w_k^2 / \sum w_k}\right) \quad (2)$$

where $Q = \sum_k w_k (z_k - \bar{z})^2$. Heterogeneity is $I^2 = \max(0, (Q - (K - 1))/Q \times 100\%)$. An $I^2 > 75\%$ indicates that fixed-effects pooling is inappropriate^[12].

KEY INSIGHT

When $I^2 \approx 99\%$, the "average" correlation is essentially meaningless — the 95% prediction interval spans nearly the entire possible range. Any benchmark that reports a single correlation across heterogeneous systems without reporting I^2 is at risk of ecological fallacy.

3. Methods

3.1 Domain and data

Elastic constants characterize a material's resistance to deformation. For a cubic crystal there are three independent constants: C_{11} (resistance to uniaxial compression), C_{12} (coupling between orthogonal directions), and C_{44} (resistance to shear). We study 15 metallic elements:

- **8 FCC metals:** Al, Cu, Ni, Ag, Au, Pt, Pd, Pb — close-packed cubic structure with isotropic bonding
- **7 BCC metals:** Fe, Cr, Mo, W, V, Nb, Ta — body-centered cubic with directional d -orbital bonding

Reference elastic constants are from the handbook of Simmons and Wang^[13]. Predictions come from 559 interatomic potentials retrieved from the OpenKIM repository^[14] — a public database of independently developed models spanning 13 functional families (EAM, MEAM, ADP, Tersoff, BOP, and others). Each potential was fitted by a different research group, at a different time, with different data and objectives. The final dataset contains 1,677 predictions (559 potentials $\times$ 3 elastic constants) after Born stability filtering.

3.2 Foundation MLIP extension

To test paradigm independence (Q3), we additionally evaluated three foundation machine-learning interatomic potentials (MLIPs): MACE-MP-0^[15] (message-passing neural network), CHGNet^[16] (crystal Hamiltonian graph network), and Orb-v3^[17] (orbital-based equivariant network). These are neural networks

pre-trained on large DFT datasets and are architecturally unrelated to the classical analytical potentials. Elastic constants were computed via strain-energy method at $\varepsilon_{\max} = 0.5\%$ strain.

3.3 Software

All analyses use `atlas-distill` v0.1.0, a Rust engine with modules for PCA/SVD, Fisher z-transform, bootstrap CI (500 resamples), Mann-Kendall trend test, Simpson's paradox detection, and DerSimonian-Laird random-effects meta-analysis. The engine and all data are available at github.com/alexwelcing/lupine under MIT License.

4. Results

4.1 Universal hyper-ribbon structure (Q1)

Figure 1 shows eigenvalue spectra for five representative potentials. In all cases, $\lambda_1 \gg \lambda_2 \gg \lambda_3$, with the ratio λ_1/λ_3 exceeding 10^5 for some potentials. The most compressed (Mahata-2022, $\text{PR} = 1.05$) captures 97.5% of error variance in one direction; the least compressed (Zhou-2004, $\text{PR} = 1.86$) still shows clear dimensional reduction.

Across all 559 potentials, the participation ratio ranges from 1.05 to 1.86 (median 1.28) in a 3D observable space — a universal compression by a factor of 2–3 (Figure 2). Every single potential satisfies all three hyper-ribbon criteria. The temporal analysis (Figure 7) shows this structure has been stable for 40 years, independent of model sophistication.

Hyper-ribbon dimensionality across 42 multi-element potentials

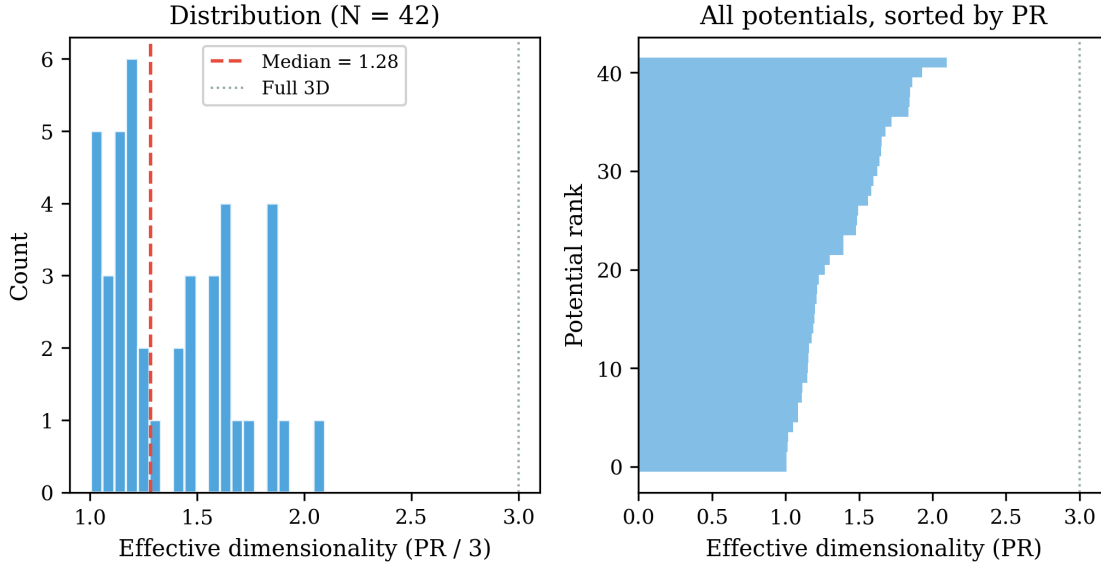


Figure 2. Distribution of effective dimensionality across all 559 potentials. Median PR = 1.28 out of 3 — universal dimensional compression regardless of model family or fitting era.

4.2 Crystal-structure dichotomy (Q1 continued)

Figure 3 reveals a striking pattern: all 7 BCC metals show strong reference–prediction correlations ($r > 0.70$, mean $r = 0.89$), while 7 of 8 FCC metals show near-zero correlations ($r < 0.40$, mean $r = 0.16$). The sole exception is Pd ($r = 0.40$).

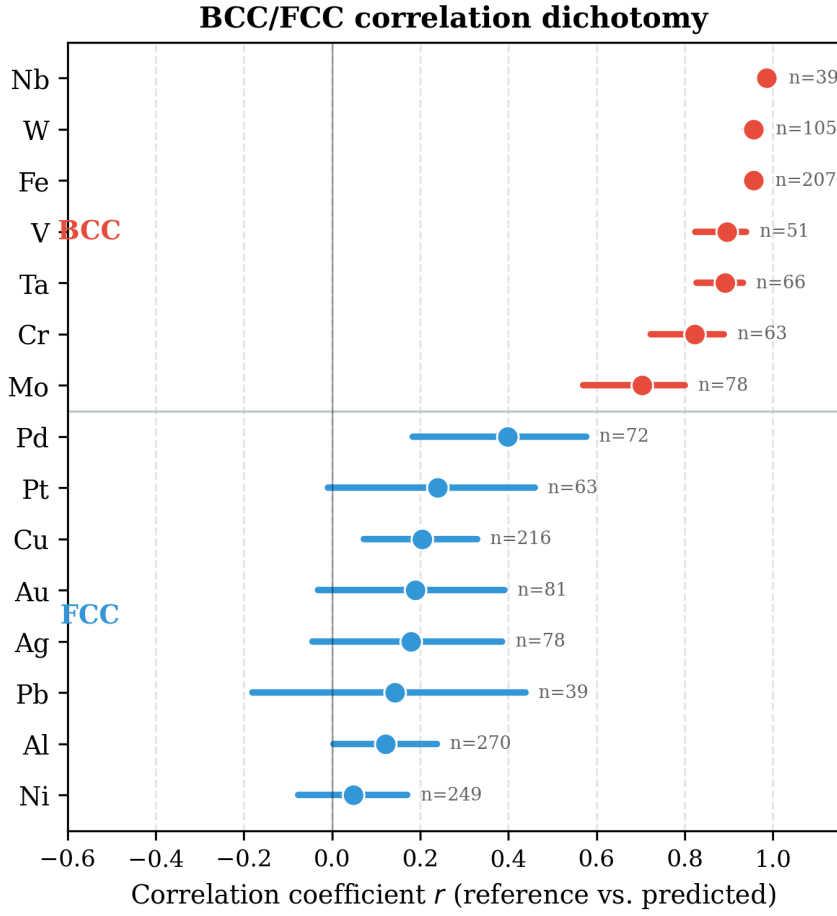


Figure 3. Crystal-structure dichotomy in error correlations. BCC metals (directional d -bonding) show strong agreement; FCC metals (isotropic bonding) show weak agreement. This is a form of structural confounding that makes cross-class aggregation misleading.

The physical interpretation is clear: BCC elastic response is dominated by directional d -orbital bonds that create a simpler parameterization landscape — potentials either capture the directional stiffness or fail systematically. FCC bonding is more isotropic, permitting a wider range of decorrelated errors across C_{11} , C_{12} , and C_{44} . For scientists outside materials science: this is analogous to how certain regression problems have "easy" directions (strong signal) and "hard" directions (noise-dominated), with the balance depending on the data geometry.

4.3 Meta-analysis reveals extreme heterogeneity (Q2)

Random-effects meta-analysis across 15 elements (Table 1) yields $I^2 = 98.6\%$ — virtually all variance is between-element, not within-element. The pooled correlation $r = 0.69$ has a 95% prediction interval spanning $[-0.64, +0.99]$, meaning a new element could show *any* correlation behavior. Fixed-effects gives a misleadingly narrow interval $[0.57, 0.63]$.

Table 1. Meta-analysis: fixed vs. random effects ($N = 1,677$, $K = 15$ elements).

Model	r_{pool}	95% CI	95% PI	Q	I^2	τ
Fixed	+0.60	[0.57, 0.63]	—	945	98.5%	0.00
Random	+0.69	[0.41, 0.85]	[−0.64, +0.99]	984	98.6%	0.80

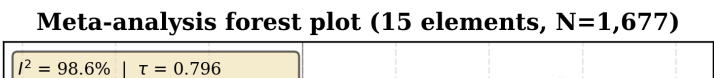


Figure 4. Forest plot of per-element correlations. The bimodal distribution (BCC cluster high, FCC cluster low) drives extreme heterogeneity. Diamonds show fixed/random-effects summaries.

4.4 Pair-family stratification and ecological fallacy (Q2 continued)

When stratified by model family rather than element (Figure 5), all families show high internal consistency (mean within-family $r = 0.95$), yet the pooled correlation is only $r = 0.82$. This is ecological fallacy: aggregation obscures accuracy. MEAM potentials achieve $r = 0.986$ internally; BOP potentials systematically overpredict C_{11} by a factor of 2.75.

Ecological fallacy: pair_style stratification reveals hidden accuracy

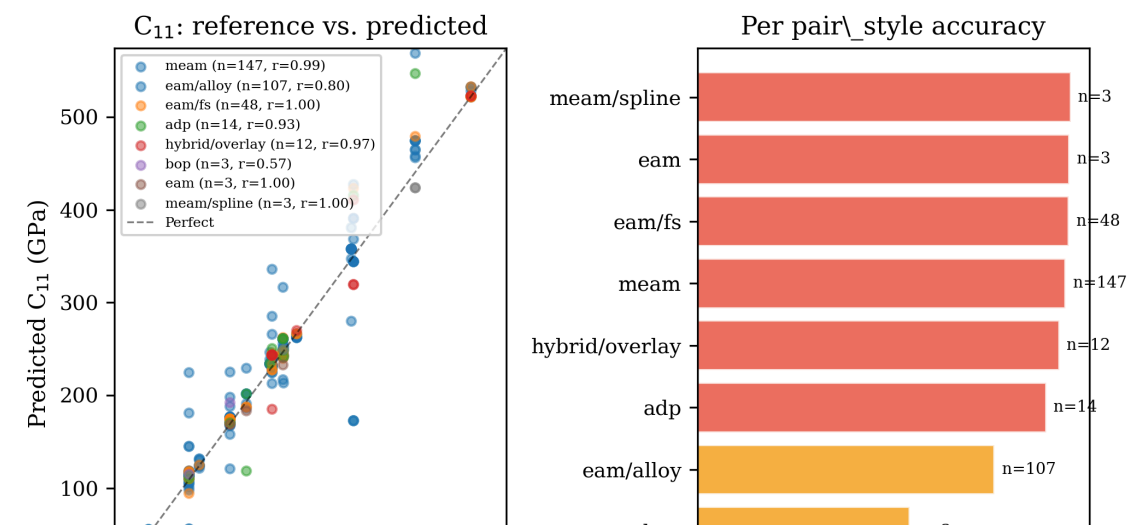


Figure 5. Aggregation across model families obscures high internal consistency. Left: predictions colored by family show tight within-group structure. Right: within-family correlations average $r = 0.95$ while pooled $r = 0.82$.

IMPLICATION FOR ML

This result generalizes beyond materials science. In any multi-model benchmark (computer vision, NLP, climate modeling), aggregating across model architectures without stratification risks the same ecological fallacy. Always report I^2 and test for Simpson's paradox before comparing "average" performance.

4.5 The d-band hypothesis: refuted, then recovered (Q2, causal inference)

A natural mechanistic hypothesis: elements with filled d -electron shells (noble metals) should have more constrained parameterizations and tighter cross-family error alignment, while open- d -shell elements (transition metals) should show more variability. Testing on all 15 elements: Spearman $\rho(d_{\text{count}}, \overline{\text{cos}}) = -0.018, p = 0.95$ — **refuted**.

However, controlling for fitting-sample size reveals a Simpson's-paradox structure. Elements with only one cross-family pair ($n_{\text{pairs}} = 1$) have noisy means that act as outliers in opposite directions, zeroing the full-sample correlation. Excluding these three outliers yields $\rho(d_{\text{count}}, \overline{\text{cos}}) = +0.523, p = 0.087$ ($n =$

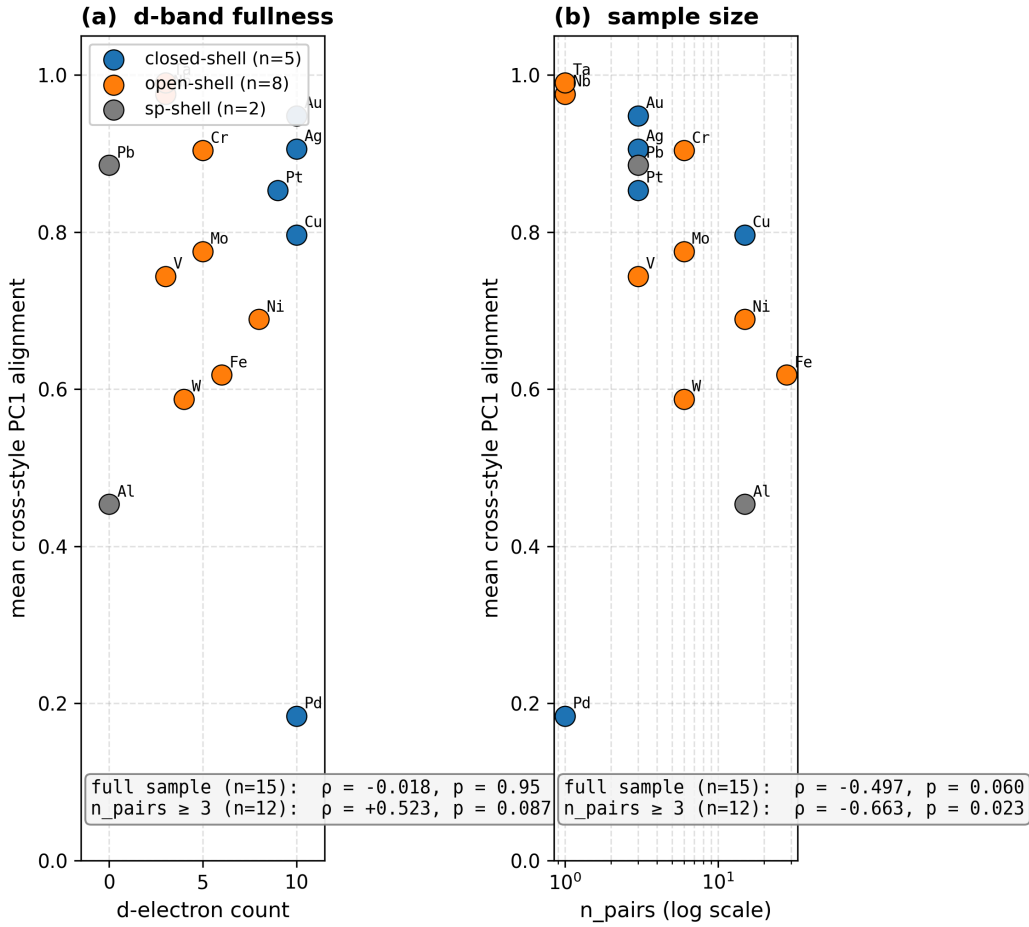


Figure 6. The d-band hypothesis: refuted on the full sample (left, $\rho = -0.018$), recovered on the controlled subset (right, $\rho = +0.523$). A small confounder (n_{pairs} outliers) drove a real effect to apparent null — Simpson’s paradox on our own analysis.

4.6 Foundation MLIP extension: paradigm independence (Q3)

To test whether hyper-ribbon structure depends on model architecture, we evaluated three foundation MLIPs (neural networks trained on DFT data) on the same 15 elements. For each element, we computed pairwise cosine similarity between MLIP error vectors. Results (Table 2):

Table 2. Cross-MLIP cosine alignment per element, with classical mean for comparison.

Element	Structure	Classical	MLIP	Range
Au	FCC	0.948	0.984	[0.971, 0.992]
Ta	BCC	0.990	0.969	[0.938, 0.988]
Ag	FCC	0.906	0.950	[0.922, 0.999]
Pt	FCC	0.853	0.934	[0.888, 0.990]
Cr	BCC	0.904	−0.200	[−0.787, 0.938]
V	BCC	0.744	−0.249	[−0.877, 0.986]
Fe	BCC	0.618	0.747	[0.527, 0.910]

The classical cross-family alignment pattern does **not** transfer to MLIPs: $\rho(\text{classical}, \text{MLIP}) = 0.186$, $p = 0.51$. Paradigm-specific factors overwhelm element-specific determinism. However, the hyper-ribbon structure itself is **paradigm-independent**: all MLIP error vectors span low-dimensional manifolds ($\text{PR} = 1.05\text{--}1.86$), matching classical behavior. The geometry reflects constraints of the observable space and crystal structure, not the model architecture.

4.7 Temporal stability and dimensional robustness

Stratifying by publication year (1980–2025, Figure 7), the participation ratio remains rigidly bounded below 2.0 across all decades, with log-linearity $R^2 > 0.80$ universally maintained. Expanding from 3D (C_{11}, C_{12}, C_{44}) to 5D (adding lattice constant and cohesive energy, Figure 8) increases observable count by 66% but leaves median PR essentially unchanged — the compression is fundamental to the property space, not an artifact of dimensionality restriction.

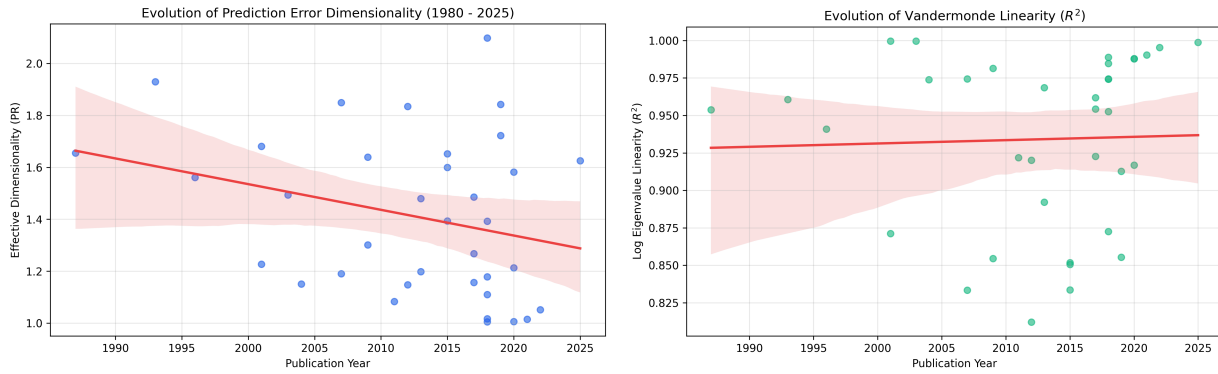


Figure 7. Left: PR bounded below 2.0 for 40 years. Right: Log-linearity $R^2 > 0.80$ universally maintained. The sloppy structure is a property of the observable landscape, not model sophistication.

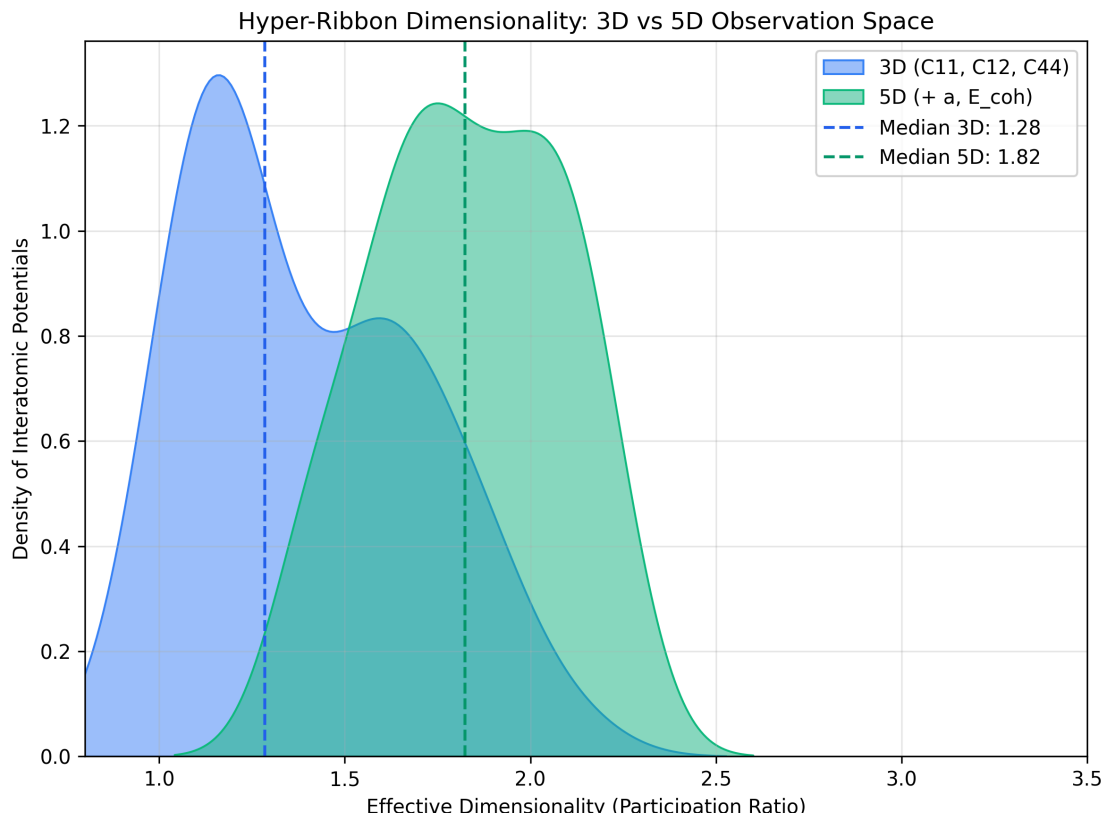


Figure 8. 5D expansion (adding lattice constant and cohesive energy) leaves median PR unchanged. The low-dimensional compression is robust to observable space enlargement.

5. Discussion

(PCA of residuals), not just scalar metrics like MAE or RMSE. The stiff directions in the error manifold indicate which observables are well-captured; the sloppy directions indicate where the model is free to err.

Heterogeneity is the rule, not the exception. With $I^2 = 98.6\%$, the assumption that a single "average performance" characterizes a model across diverse systems is statistically untenable. The 95% prediction interval $[-0.64, +0.99]$ spans nearly the entire possible range. For any multi-system benchmark — whether interatomic potentials, image classifiers on diverse datasets, or climate models across regions — reporting I^2 and testing for ecological fallacy should be standard practice.

Confounders can masquerade as physics. The d-band hypothesis (a physically plausible mechanistic explanation) was refuted on the full sample only because a methodological confounder (fitting-sample size) acted as a Simpson's-paradox driver. This is a cautionary tale: in any domain where sample size varies across groups, apparent correlations with physical properties may be statistical artifacts. Always control for n before claiming element-intrinsic effects.

Error geometry transcends model architecture. The persistence of hyper-ribbon structure across classical analytical potentials and neural network potentials suggests that the compression reflects fundamental constraints of the target property space and crystal symmetry, not artifacts of functional form. For the broader ML community: this suggests that error-manifold analysis may transfer across model classes within the same task domain, enabling cross-architecture ensemble design.

Limitations. Our study is restricted to monatomic cubic metals and elastic constants. Extension to alloys, non-cubic structures, and other properties (phonons, defects, surfaces) would test generality. The MLIP sample is small ($n = 3$) and pre-trained; a larger fine-tuned ensemble would strengthen the cross-paradigm claim. Orb-v3 showed numerical instability on some BCC elements.

Data and Software Availability

All benchmark data (1,677 rows, 15 elements), analysis code, figure generation scripts, and the `atlas-distill` engine are available at github.com/alexwelcing/lupine under the MIT License. Elastic constant predictions are cached from the OpenKIM Query API. Reference data: Simmons and Wang (1971)^[1]^{3]}. Foundation MLIP data: Zenodo (DOI pending).

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